

MAT41-2

Advanced Estimation

Method of moments, maximum likelihood, and the asymptotic theory of estimators

Not just how to build an estimator, but how to prove it is good — and how good it can possibly be.

Albert School — 26 hours

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Preface

In **Probability & Statistics I** (MAT21-2) you met estimation for the first time. You learned that an **estimator** is any rule $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$ that turns data into a guess at an unknown parameter, you judged such rules by their **bias** and **mean squared error**, and you met a principled recipe for manufacturing them — **maximum likelihood**: write the likelihood $L(\theta) = \prod_i p(x_i; \theta)$, take its logarithm, differentiate, set the derivative to zero, and solve. You ran this machine on the Bernoulli, binomial and Poisson distributions and recovered estimators that felt inevitable — the sample proportion, the sample mean.

But every one of those models was **discrete**: the likelihood was a product of **probabilities**. Most measured quantities are not counts. A failure time, a concentration, a measurement error vary **continuously**, and in MAT31-2 you learned to describe them with **densities** rather than mass functions, met the uniform, exponential, normal, chi-squared and Student-*t* families, and — crucially — proved the two theorems that this entire course rests on: the **Law of Large Numbers**, which says sample averages settle onto their expectations, and the **Central Limit Theorem**, which says they do so with a **normal** shape at rate $1/\sqrt{n}$. You used the CLT to build your first **confidence intervals**.

This course completes the story in three movements. First, we carry estimation across the discrete–continuous divide: the **method of moments** and **maximum likelihood for densities**, where the likelihood becomes a product of density **values** and the calculus of MAT11-1

does the heavy lifting. Second, we ask the question MAT21-2 could only pose informally — **which estimator is best?** — and answer it with bias, MSE, efficiency, and the **bias–variance tradeoff**. Third, and most deeply, we study what happens **as the sample grows**. We make precise what it means for an estimator to converge (**convergence in probability, almost sure convergence, consistency**), we state the remarkable fact that the MLE is **asymptotically normal** with a variance governed by a single quantity — the **Fisher information** — and we discover, through the **Cramér–Rao bound**, that there is a hard floor below which no unbiased estimator’s variance can fall. The MLE, asymptotically, sits exactly on that floor.

The method is the one you know: every idea arrives as a **concrete example first** and the **general statement second**, we name the mistakes students reliably make, and we never lose sight of the question behind the algebra — **why should we believe this estimate, and how wrong might it be?**

1 The method of moments

Maximum likelihood, for all its power, can be analytically painful: the score equation may have no closed-form solution. The **method of moments** (MoM) is an older, cruder, but wonderfully direct alternative. Its idea is almost naïve, and that is its charm: **the sample should look like the population it came from**, so choose the parameter values that make the model’s theoretical moments equal the moments actually observed in the data.

1.1 One parameter: matching the mean

Worked example — the rate of a Poisson-like waiting process. Service times at a help desk are modelled as $X \sim \text{Exp}(\lambda)$, and we have recorded n of them: $x_1, \dots, x_n$. The exponential’s theoretical mean is $\mathbb{E}[X] = 1/\lambda$ (MAT31-2). The data’s **sample mean** is $\bar{x} = (1/n) \sum_i x_i$. The method of moments simply **equates the two**:

$$\frac{1}{\lambda} = \bar{x} \implies \hat{\lambda} = \frac{1}{\bar{x}}.$$

If the average service time is 4 minutes, estimate the rate as 1/4 per minute. No likelihood, no derivative — just “make the model’s mean match the data’s mean”.

The recipe in its simplest form: the **first theoretical moment** $m_1(\theta) = \mathbb{E}_\theta[X]$ is a known function of the unknown parameter; the **first sample moment** is $\bar{X} = (1/n) \sum_i X_i$; set them equal and solve for θ .

Definition 1 (sample moments) The **k -th sample moment** of data $X_1, \dots, X_n$ is

$$\overline{X^k} = \frac{1}{n} \sum_{i=1}^n X_i^k.$$

In particular the first sample moment is the sample mean $\bar{X}$, and the second is $\overline{X^2} = (1/n) \sum_i X_i^2$.

Why is matching moments a legitimate thing to do? Because of the Law of Large Numbers. By the LLN, $\bar{X}^k \rightarrow \mathbb{E}[X^k]$ as $n \rightarrow \infty$: the sample moments converge to the true theoretical moments. So forcing them equal is forcing the estimate toward the value that **would** be exactly right with infinite data. The method of moments is the LLN turned into an estimation recipe.

1.2 Several parameters: matching several moments

A model with p unknown parameters needs p equations, so we match the first p moments: $m_1(\theta) = \bar{X}$, $m_2(\theta) = \overline{X^2}$, and so on, then solve the resulting system.

Worked example — both parameters of a normal. Suppose $X \sim \mathcal{N}(\mu, \sigma^2)$ with **both** parameters unknown. Two unknowns require two moment equations. The theoretical moments are

$$m_1 = \mathbb{E}[X] = \mu, \quad m_2 = \mathbb{E}[X^2] = \text{Var}(X) + (\mathbb{E}[X])^2 = \sigma^2 + \mu^2,$$

using the König–Huygens identity in reverse. Matching:

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 + \hat{\mu}^2 = \overline{X^2} \implies \hat{\sigma}^2 = \overline{X^2} - \bar{X}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

The MoM estimators are the **sample mean** and the **sample variance** (with divisor n) — reassuringly the obvious choices.

Worked example — the upper end of a uniform. A sensor returns readings $X \sim \text{Unif}(0, \theta)$ with θ unknown. The theoretical mean is $\mathbb{E}[X] = \theta/2$. Matching to the sample mean:

$$\frac{\theta}{2} = \bar{X} \implies \hat{\theta}_{\text{MoM}} = 2\bar{X}.$$

This is delightfully simple — “the maximum is about twice the average” — and we will compare it head-to-head with the maximum-likelihood estimator later, with a surprising result.

Common pitfall The method of moments can return **nonsensical** estimates — values outside the parameter’s legal range. In the uniform example, if the data happen to be 0.1, 0.2, 0.9, then $\bar{X} = 0.4$ and $\hat{\theta} = 0.8$ — but we **observed** a value $0.9 > 0.8$, which is impossible under $\text{Unif}(0, 0.8)$. Matching moments uses only the average and ignores that the largest observation is itself a lower bound for θ . This is the price of the method’s crudeness: it is consistent (right in the limit) but can be embarrassing on a single sample.

Remark When a choice of moments is available, prefer the **lowest** ones. Higher sample moments ($\bar{X}^3$, $\bar{X}^4$, ...) have larger variance — they are dominated by rare extreme observations — so estimators built from them are noisier. Match the mean before the variance, the variance before the skewness.

Exercises

1. For $X \sim \text{Exp}(\lambda)$, derive the MoM estimator of λ using the **second** moment $\mathbb{E}[X^2] = 2/\lambda^2$ instead of the first. Is it the same as $1/\bar{X}$?
2. For the gamma density with known shape k and unknown rate β , where $\mathbb{E}[X] = k/\beta$, find the MoM estimator of β .
3. $X \sim \text{Unif}(a, b)$ with both endpoints unknown. Using $\mathbb{E}[X] = (a + b)/2$ and $\text{Var}(X) = (b - a)^2/12$, derive MoM estimators for a and b .

2 Maximum likelihood for continuous densities

We now carry maximum likelihood across the divide. The logic is **identical** to the discrete case of MAT21-2 — choose the parameter that makes the observed data most plausible — but the likelihood is now a product of density **values** $f(x_i; \theta)$ rather than probabilities. There is a subtlety in interpretation (densities are not probabilities) and none at all in the mechanics.

2.1 The likelihood of a continuous sample

Worked example — which rate best explains these failure times?. Three components fail at times $x_1 = 1$, $x_2 = 2$, $x_3 = 4$ hours, modelled as i.i.d. $\text{Exp}(\lambda)$. For a candidate rate λ , the **joint density** at the observed data is

$$f(x_1; \lambda)f(x_2; \lambda)f(x_3; \lambda) = (\lambda e^{-\lambda})(\lambda e^{-2\lambda})(\lambda e^{-4\lambda}) = \lambda^3 e^{-7\lambda}.$$

Read as a function of λ , this is the **likelihood** $L(\lambda)$: a measure of how well each rate explains the data. A tiny λ makes long waits likely but this product small; a huge λ makes the early failures likely but the late one almost impossible. Somewhere in between sits the rate that maximises the product — that is the MLE.

Definition 2 (likelihood and log-likelihood for a density) For i.i.d. observations $x_1, \dots, x_n$ from a density $f(x; \theta)$, the **likelihood** is the joint density viewed as a function of θ :

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta),$$

and the **log-likelihood** is $\ell(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln f(x_i; \theta)$. The **maximum likelihood estimator** $\hat{\theta}$ is the value maximising L — equivalently ℓ , since $\ln$ is increasing.

Common pitfall A continuous likelihood is a product of **densities**, not probabilities, and can be **larger than 1** (a sharp density easily exceeds 1). This is harmless: the MLE cares only about **where** L is largest, not about its absolute size. Do not be alarmed by a “likelihood of 37”. What would be wrong is to call it a probability.

As in MAT21-2 we always work with ℓ , not L : the logarithm turns the awkward **product** into a **sum**, whose derivative is the sum of derivatives. The recipe is the three-step ritual you already know — now applied to densities.

2.2 The score equation and the second-order condition

Definition 3 (score equation) The **score** is the derivative of the log-likelihood, $\ell'(\theta) = \frac{\partial}{\partial \theta} \sum_i \ln f(x_i; \theta)$. The MLE solves the **score equation**

$$\ell'(\theta) = 0,$$

and is confirmed to be a maximum (not a minimum or saddle) by the **second-order condition** $\ell''(\hat{\theta}) < 0$ — the log-likelihood is concave down at the peak.

Worked example — MLE for the exponential rate. For $X \sim \text{Exp}(\lambda)$, $f(x; \lambda) = \lambda e^{-\lambda x}$, so

$$\ell(\lambda) = \sum_{i=1}^n (\ln \lambda - \lambda x_i) = n \ln \lambda - \lambda \sum_{i=1}^n x_i.$$

The score equation:

$$\ell'(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0 \implies \hat{\lambda} = \frac{n}{\sum_i x_i} = \frac{1}{\bar{X}}.$$

The second-order condition holds, $\ell''(\lambda) = -n/\lambda^2 < 0$, so this is genuinely the maximum. For the failure times 1, 2, 4 of the opening example, $\hat{\lambda} = 3/7 \approx 0.43$ per hour — exactly the peak of $\lambda^3 e^{-7\lambda}$. Note the MLE **coincides** with the method-of-moments estimator here; that is a happy accident of the exponential, not a general rule.

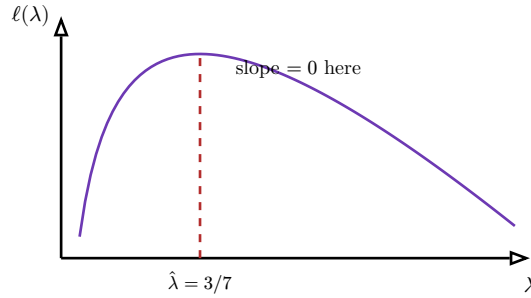


Figure 1: The log-likelihood $\ell(\lambda) = 3 \ln \lambda - 7\lambda$ for the failure times 1, 2, 4. It rises, peaks where the score $\ell'(\lambda) = 0$, then falls — concave throughout ($\ell'' < 0$), so the stationary point is the unique maximum. The MLE $\hat{\lambda} = 3/7$ is the λ at the peak.

Worked example — MLE for the normal — both parameters. For $X \sim \mathcal{N}(\mu, \sigma^2)$ with density $f(x) = (1/(\sigma\sqrt{2\pi}))e^{-(x-\mu)^2/(2\sigma^2)}$,

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

This has **two** parameters, so we set **both** partial derivatives to zero. The μ -score:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_i (x_i - \mu) = 0 \implies \hat{\mu} = \bar{X}.$$

The σ^2 -score:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (x_i - \mu)^2 = 0 \implies \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

The MLE recovers the sample mean and the sample variance **with divisor n** — identical to the method of moments here, and, as we will see in the next chapter, **biased** for σ^2 .

2.3 When the score equation fails

Common pitfall The score-equation recipe finds **interior** maxima — points where the derivative vanishes. It is **blind to maxima on the boundary** of the parameter range, and to densities that are not differentiable in θ . The uniform distribution is the canonical trap.

Worked example — MLE for the uniform upper end — a boundary maximum. For $X \sim \text{Unif}(0, \theta)$, the density is $f(x; \theta) = 1/\theta$ for $0 \leq x \leq \theta$ and 0 otherwise. The likelihood is

$$L(\theta) = \prod_{i=1}^n \frac{1}{\theta} \cdot \mathbb{1}\{0 \leq x_i \leq \theta\} = \frac{1}{\theta^n} \quad \text{provided } \theta \geq \max_i x_i, \quad \text{else } 0.$$

Differentiating $1/\theta^n$ gives $-n/\theta^{n+1}$, which is **never zero** — the score equation has no solution! But the likelihood is decreasing in θ wherever it is positive, and it is positive only for $\theta \geq \max_i x_i$. So L is largest at the **smallest admissible** θ :

$$\hat{\theta}_{\text{MLE}} = \max(X_1, \dots, X_n).$$

The MLE is the largest observation. Calculus alone would have missed it entirely; you must reason about the **shape** of L , not just its derivative.

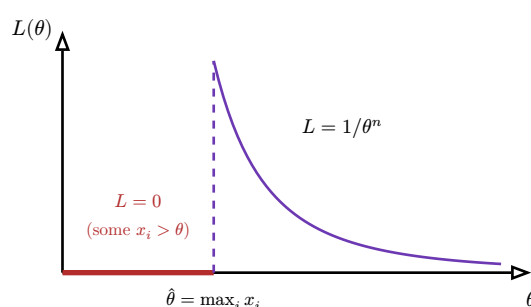


Figure 2: The uniform likelihood $L(\theta)$ for $\text{Unif}(0, \theta)$. It is zero until θ reaches the largest observation (any smaller θ cannot have produced that data point), jumps to its maximum there, then decays as $1/\theta^n$. The maximum sits on the **boundary** $\theta = \max_i x_i$, where the derivative is not zero — the score equation never finds it.

Exercises

4. Derive the MLE of λ for $X \sim \text{Exp}(\lambda)$ from a sample, including the second-order check, and confirm it equals $1/\bar{X}$.
5. For the Pareto density $f(x; \alpha) = \alpha/x^{\alpha+1}$ on $x \geq 1$, write the log-likelihood and derive the MLE $\hat{\alpha}$.
6. For $X \sim \mathcal{N}(\mu, \sigma^2)$ with σ^2 **known**, show the MLE of μ is $\bar{X}$ and that the log-likelihood is exactly a downward parabola in μ .
7. Explain in your own words why the score equation fails for $\text{Unif}(0, \theta)$ but succeeds for $\text{Exp}(\lambda)$.

3 Comparing estimators

We now have **two** recipes — moments and likelihood — and they sometimes disagree (the uniform: $2\bar{X}$ versus $\max_i X_i$). Which is better? To answer we need a vocabulary for **estimator quality**. MAT21-2 introduced bias and MSE; here we sharpen them, add **efficiency**, and make the **bias–variance tradeoff** precise.

3.1 Bias, variance, and mean squared error

Recall from MAT21-2 that an estimator $\hat{\theta}$ is a random variable (it depends on the random sample), so it has its own distribution — the **sampling distribution** — with a mean and a variance.

Definition 4 (bias, variance, MSE) For an estimator $\hat{\theta}$ of a parameter θ ,

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta, \quad \text{Var}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2], \quad \text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2].$$

The estimator is **unbiased** if $\text{Bias}(\hat{\theta}) = 0$. The MSE measures total average squared error from the **true** target θ .

The single most important identity in estimation theory ties these together. It was stated in MAT21-2; we restate it because everything in this chapter leans on it.

Theorem 5 (bias–variance decomposition)

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2.$$

Proof. Write $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}[\hat{\theta}]) + (\mathbb{E}[\hat{\theta}] - \theta)$. The first bracket has mean zero; the second is the constant bias. Square and take expectations: the cross term $2\mathbb{E}[\hat{\theta} - \mathbb{E}[\hat{\theta}]] \cdot \text{Bias}$ vanishes, leaving $\text{Var}(\hat{\theta}) + (\text{Bias})^2$. $\square$

Total error decomposes into two independent failures: **systematic** error (bias — aiming at the wrong place) and **random** error (variance — scattering around wherever you aim). The dartboard picture below is the image to carry through the rest of the course.

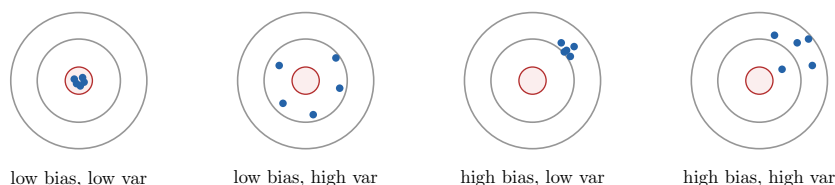


Figure 3: Bias versus variance. Bias is how far the **cluster's centre** sits from the bullseye (the true θ); variance is how **spread out** the shots are. MSE combines both: the best estimator (far left) is centred **and** tight. An unbiased but high-variance estimator (second) can be worse on a single sample than a slightly biased but tight one.

3.2 Efficiency: comparing two unbiased estimators

When two estimators are both unbiased, MSE reduces to variance, and the comparison is clean: **prefer the one with smaller variance**. This is the notion of **efficiency**.

Definition 6 (relative efficiency) For two unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ of θ , the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}.$$

A value greater than 1 means $\hat{\theta}_1$ is more efficient (tighter). The **more efficient** estimator extracts more information from the same data.

Worked example — two estimators of the uniform's upper end. Return to $X \sim \text{Unif}(0, \theta)$ and its two estimators.

The **method of moments** gave $\hat{\theta}_1 = 2\bar{X}$. It is unbiased: $\mathbb{E}[2\bar{X}] = 2 \cdot \theta/2 = \theta$. Its variance, using $\text{Var}(X) = \theta^2/12$ and variance-of-an-average, is

$$\text{Var}(\hat{\theta}_1) = 4 \cdot \text{Var}(\bar{X}) = 4 \cdot \frac{\theta^2}{12n} = \frac{\theta^2}{3n}.$$

The **maximum likelihood** gave $\hat{\theta}_2 = \max_i X_i =: X_{(n)}$. One can show $\mathbb{E}[X_{(n)}] = \frac{n}{n+1}\theta$, so the MLE is **biased downward** (it can never exceed the true θ). The **bias-corrected** version $\hat{\theta}_3 = \frac{n+1}{n}X_{(n)}$ is unbiased, with variance

$$\text{Var}(\hat{\theta}_3) = \frac{\theta^2}{n(n+2)}.$$

Now compare the two unbiased estimators:

$$\text{eff}(\hat{\theta}_3, \hat{\theta}_1) = \frac{\text{Var}(\hat{\theta}_1)}{\text{Var}(\hat{\theta}_3)} = \frac{\theta^2/(3n)}{\theta^2/(n(n+2))} = \frac{n+2}{3}.$$

For $n = 10$ this is 4; for $n = 100$ it is 34. The maximum-based estimator is **dramatically** more efficient — its variance shrinks like $1/n^2$, not $1/n$. The reason is intuitive: the largest observation carries far more information about an upper limit than the average does.

Common pitfall Unbiasedness is not the goal; small MSE is. The raw MLE $X_{(n)}$ is biased, yet its MSE can still beat the unbiased $2\bar{X}$. Chasing unbiasedness for its own sake — “correct on average” — can lead you to a worse estimator. Always ask about **total error** (MSE), not bias alone.

3.3 The bias–variance tradeoff

The dartboard shows that bias and variance are **different** failures; the **tradeoff** is the discovery that they are often **in tension** — that you can **buy** a reduction in one by paying in the other, and that the MSE-minimising choice is usually **not** at zero bias.

Worked example — estimating the normal variance — the n versus $n-1$ question. For $\mathcal{N}(\mu, \sigma^2)$, consider the family of variance estimators

$$S_c^2 = \frac{1}{c} \sum_{i=1}^n (X_i - \bar{X})^2$$

for different divisors c . Three choices matter:

- $c = n - 1$: the **sample variance**, which is **unbiased** ($\mathbb{E}[S_{n-1}^2] = \sigma^2$);
- $c = n$: the **MLE**, biased low by a factor $(n - 1)/n$;
- $c = n + 1$: minimises the MSE among this family.

The MSE-optimal divisor is **larger** than $n - 1$: accepting a little downward bias **reduces variance enough** to lower total error. The famous “ $n - 1$ ” corrects the bias but is **not** MSE-optimal — a concrete illustration that the best estimator deliberately tolerates bias.

Remark The bias–variance tradeoff is not a curiosity of variance estimation — it is the organising principle of **all** of statistical learning. Every regularised model (ridge regression,

shrinkage, smoothing, early stopping) deliberately **injects bias** to **cut variance** and so lower prediction error. The “ $n + 1$ beats $n - 1$ ” example is the simplest instance of the idea that will dominate your machine-learning courses.

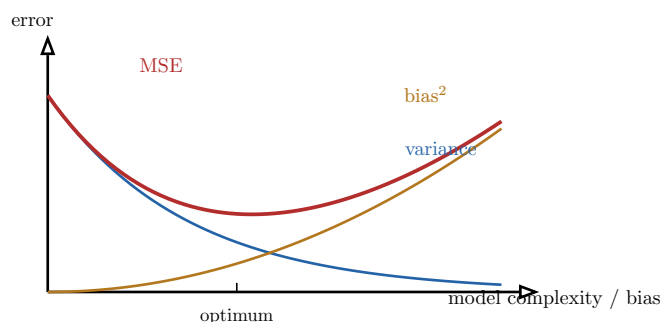


Figure 4: The bias–variance tradeoff. As an estimator is made more flexible, its variance falls but its (squared) bias rises. Their sum, the MSE, is minimised at an **interior** point — neither the zero-bias extreme nor the zero-variance extreme. The lowest-error estimator deliberately accepts some bias.

Exercises

8. Show $\mathbb{E}\left[(1/n) \sum_i (X_i - \bar{X})^2\right] = (n-1)/n \cdot \sigma^2$, hence the MLE variance is biased low, and the divisor $n-1$ fixes it.
9. Two unbiased estimators of μ have variances σ^2/n and $1.4\sigma^2/n$. Which is more efficient, and what is the relative efficiency?
10. For $\text{Unif}(0, \theta)$, verify $\mathbb{E}[\max_i X_i] = (n/(n+1))\theta$ by integrating against the density nx^{n-1}/θ^n of the maximum.

4 Convergence and consistency

So far “good for large n ” has been informal. The asymptotic theory that follows — the heart of this course — demands precision: **in what sense** does an estimator approach the truth as data accumulate? There is more than one answer, and the distinctions matter. We make two modes of convergence precise and define the property every usable estimator must have: **consistency**.

4.1 Convergence in probability

Worked example — the sample mean closing in. Let $\bar{X}_n$ be the running sample mean of i.i.d. observations with mean μ and variance σ^2 . Fix a tolerance, say $\varepsilon = 0.01$. The probability that $\bar{X}_n$ misses μ by more than 0.01 is, by Chebyshev, $P(|\bar{X}_n - \mu| > 0.01) \leq \sigma^2/(n \cdot 0.01^2)$, which $\rightarrow 0$ as $n \rightarrow \infty$. **For any tolerance you name**, the chance of exceeding it eventually becomes negligible. This is convergence in probability.

Definition 7 (convergence in probability) A sequence of random variables Y_n **converges in probability** to a constant c , written $Y_n \xrightarrow{P} c$, if for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|Y_n - c| > \varepsilon) = 0.$$

This is exactly the language in which the **Weak Law of Large Numbers** of MAT31-2 was stated: $\bar{X}_n \xrightarrow{P} \mu$. Convergence in probability says the **probability of a non-trivial error** shrinks

to zero. It does **not** promise that any particular sequence of outcomes settles down — only that, at each large n , being far off is unlikely.

4.2 Almost sure convergence

Worked example — watching a single run forever. Imagine fixing one infinite sequence of coin flips and plotting the running proportion of heads $\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots$ for **that one experiment**. Almost sure convergence is the statement that **this very path** settles onto $1/2$ and stays there — not just that it is **probably** close at each n , but that the whole trajectory has a limit equal to $1/2$. The **Strong Law of Large Numbers** guarantees this for all but a negligible (probability-zero) set of exceptional sequences.

Definition 8 (almost sure convergence) Y_n converges almost surely to c , written $Y_n \xrightarrow{\text{a.s.}} c$, if

$$P\left(\lim_{n \rightarrow \infty} Y_n = c\right) = 1.$$

That is, the **sequence itself** converges to c with probability 1.

The grammatical position of the limit is the whole difference. In **convergence in probability** the limit is **outside**: $\lim_n P(\dots) = 0$ — a statement about a sequence of probabilities. In **almost sure convergence** the limit is **inside**: $P(\lim_n \dots) = 1$ — a statement about the limit of the sequence of values. Almost sure convergence is **stronger**: it implies convergence in probability, but not conversely.

Common pitfall The two modes are genuinely different, and the difference is not pedantry. Under convergence in probability, $\bar{X}_n$ may **exceed** the tolerance infinitely often — just at an ever-thinning set of n 's — while still converging in probability. Almost sure convergence forbids this: past some (random) point the path **never** strays again. For estimating a parameter from a **fixed, growing** dataset, almost sure convergence is the honest description of what “the estimate settles down” means. The Weak Law gives $\xrightarrow{P}$; the Strong Law gives $\xrightarrow{\text{a.s.}}$.

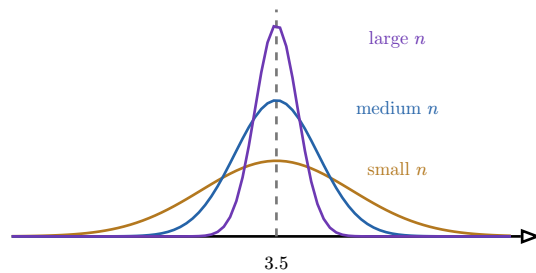


Figure 5: Consistency as concentration. As n grows, the sampling distribution of a consistent estimator $\hat{\theta}$ piles up ever more tightly around the true value θ : the bias vanishes and the variance collapses. In the limit all the probability mass sits at θ .

4.3 Consistency

Definition 9 (consistency) An estimator $\hat{\theta}_n$ is **consistent** for θ if

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty,$$

for every true value of θ . (If the convergence is almost sure, the estimator is **strongly consistent**.)

Consistency is the **minimum** requirement for an estimator to be taken seriously: **with enough data, it gets arbitrarily close to the truth**. An inconsistent estimator is broken — more data does not help it.

Proposition 10 (a sufficient condition for consistency) If $\hat{\theta}_n$ is **asymptotically unbiased** ($\mathbb{E}[\hat{\theta}_n] \rightarrow \theta$) and its **variance vanishes** ($\text{Var}(\hat{\theta}_n) \rightarrow 0$), then it is consistent.

Proof. Since $\text{MSE}(\hat{\theta}_n) = \text{Var}(\hat{\theta}_n) + (\text{Bias}(\hat{\theta}_n))^2 \rightarrow 0$, Chebyshev's inequality gives $P(|\hat{\theta}_n - \theta| > \varepsilon) \leq \text{MSE}(\hat{\theta}_n)/\varepsilon^2 \rightarrow 0$. $\square$

This is the workhorse test: **bias $\rightarrow 0$ and variance $\rightarrow 0$ together imply consistency**. It instantly certifies the estimators we have built.

Worked example — consistency of the estimators we have met.

- $\bar{X}_n$ for μ : unbiased, variance $\sigma^2/n \rightarrow 0$. **Consistent**. (This is the LLN.)
- $\hat{\lambda} = 1/\bar{X}$ for the exponential rate: $\bar{X} \xrightarrow{P} 1/\lambda$ by the LLN, and $1/x$ is continuous, so $\hat{\lambda} \xrightarrow{P} \lambda$. **Consistent** (via the continuous-mapping principle — a continuous function of a consistent estimator is consistent).
- $X_{(n)} = \max_i X_i$ for the uniform's θ : biased by $\theta/(n+1) \rightarrow 0$ and variance $\rightarrow 0$. **Consistent**, despite the bias.
- The MoM and MLE **in general**: both are consistent under mild conditions, because matching sample moments (which converge to true moments) and maximising the log-likelihood (whose per-observation average converges to a function maximised at the true θ) both lock onto the truth as $n \rightarrow \infty$.

Common pitfall A biased estimator can still be consistent (the uniform maximum), and an unbiased estimator can fail to be useful if its variance does **not** vanish. Consistency is about the **limit**; unbiasedness is about **finite samples**. Do not conflate them: “is it unbiased?” and “is it consistent?” are different questions with different answers.

Exercises

11. State the difference between $\lim_n P(|Y_n - c| > \varepsilon) = 0$ and $P(\lim_n Y_n = c) = 1$ in one sentence each, and say which implies which.
12. Show the MoM estimator $\hat{\theta} = 2\bar{X}$ for $\text{Unif}(0, \theta)$ is consistent.
13. Give the consistency argument for the Poisson MLE $\hat{\lambda} = \bar{X}$.
14. Is an estimator with $\text{Var}(\hat{\theta}_n) = 1$ for all n and zero bias consistent? Explain.

5 Fisher information and the Cramér–Rao bound

We have seen estimators with wildly different variances. A profound question arises: **is there a limit to how good an estimator can be?** Is there a floor below which no unbiased estimator's variance can fall? The astonishing answer is **yes**, and the floor is set by a single quantity computed from the model alone — the **Fisher information**. This chapter is the theoretical climax: it tells us the best possible precision **before we choose any estimator**.

5.1 Fisher information: how much a sample tells you

Worked example — sharp peaks and flat peaks. Two log-likelihoods, two datasets. In one, $\ell(\theta)$ rises to a **sharp, narrow** peak: nudging θ away from $\hat{\theta}$ makes the data much less likely, so the data **pin down** θ tightly. In the other, $\ell(\theta)$ has a **broad, flat** peak: a wide range of θ values explain the data almost equally well, so θ is poorly determined. The **curvature** of the log-likelihood at its peak — how sharply it bends — measures how much **information** the data carry about θ . Fisher information is the average of that curvature.

Definition 11 (Fisher information) For a single observation from $f(x; \theta)$, the **Fisher information** is the expected curvature of the log-likelihood:

$$I(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \ln f(X; \theta) \right] = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(X; \theta) \right)^2 \right].$$

For n i.i.d. observations the information **adds**: $I_{n(\theta)} = n I(\theta)$.

The two expressions are equal under mild regularity (the score has mean zero, so its variance equals minus the expected second derivative). Use whichever is easier: the **second-derivative** form is usually the quickest to compute; the **squared-score** form makes the additivity over observations transparent.

Worked example — Fisher information of the exponential. For $f(x; \lambda) = \lambda e^{-\lambda x}$, $\ln f = \ln \lambda - \lambda x$, so $(\partial/\partial \lambda) \ln f = 1/\lambda - x$ and $(\partial^2/\partial \lambda^2) \ln f = -1/\lambda^2$. Hence

$$I(\lambda) = -\mathbb{E} \left[-\frac{1}{\lambda^2} \right] = \frac{1}{\lambda^2}.$$

Information is large when λ is small (slow processes, long informative waits) and small when λ is large. For n observations, $I_{n(\lambda)} = n/\lambda^2$.

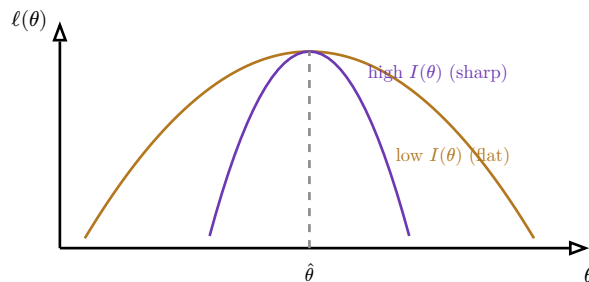


Figure 6: Fisher information is curvature at the peak. A **sharply** curved log-likelihood (purple) pins θ down precisely — high information, low estimator variance. A **flat** one (orange) leaves θ poorly determined — low information, high variance. The Cramér–Rao bound makes this inverse relationship exact.

5.2 The Cramér–Rao lower bound

Theorem 12 (Cramér–Rao bound) Under regularity conditions, **any** unbiased estimator $\hat{\theta}$ of θ based on n i.i.d. observations satisfies

$$\text{Var}(\hat{\theta}) \geq \frac{1}{n I(\theta)}.$$

The right-hand side is the **Cramér–Rao lower bound** (CRLB). An unbiased estimator attaining it is called **efficient**.

This is a remarkable statement: it bounds the variance of **every conceivable unbiased estimator** — including ones nobody has invented — using only the model. It says information and precision are **inversely** related: double the information and you can at best halve the variance. The $1/n$ is the familiar sampling rate; the $1/I(\theta)$ is the new content.

Worked example — the exponential MLE is efficient. The MLE $\hat{\lambda} = 1/\bar{X}$ has (for large n) variance λ^2/n . The Cramér–Rao bound, using $I(\lambda) = 1/\lambda^2$, is

$$\frac{1}{nI(\lambda)} = \frac{\lambda^2}{n}.$$

The MLE’s variance **equals** the bound: no unbiased estimator of λ can do better. The MLE is **efficient**. (Strictly, $1/\bar{X}$ is slightly biased in finite samples; the statement is exact asymptotically — the subject of the next chapter.)

5.3 The exponential family and efficiency

The OP singles out the **one-parameter exponential family** — the class of densities that can be written

$$f(x; \theta) = h(x) \exp(\eta(\theta) T(x) - A(\theta))$$

for some functions h, η, T, A . This family (which contains the exponential, normal-with-known-variance, Poisson, Bernoulli, gamma-with-known-shape, and more) is exactly the class for which the MLE is **efficient in finite samples** — its variance hits the Cramér–Rao bound exactly, not merely in the limit. Computing $I(\theta)$ for such a model and comparing $1/(nI(\theta))$ to the MLE’s variance is the standard test of whether an estimator can be improved upon.

Worked example — is the Bernoulli MLE efficient? The Bernoulli(p) pmf $p^x(1-p)^{1-x}$ is in the exponential family. Its score is $(\partial/\partial p) \ln f = x/p - (1-x)/(1-p)$, and one finds

$$I(p) = \frac{1}{p(1-p)}.$$

The MLE is $\hat{p} = \bar{X}$ with variance $p(1-p)/n$, while the CRLB is $1/(nI(p)) = p(1-p)/n$. They coincide: the sample proportion is **efficient** — no unbiased estimator of a probability beats it. This vindicates, with a hard theorem, the “natural” estimator you have used since MAT21-2.

Common pitfall The Cramér–Rao bound applies to **unbiased** estimators. A **biased** estimator can have variance **below** the CRLB — there is no contradiction, because the bound constrains variance only under unbiasedness. (Indeed the whole bias–variance tradeoff exploits exactly this loophole.) When you read “the MLE achieves the Cramér–Rao bound”, the unbiasedness qualifier — at least asymptotically — is doing essential work.

Exercises

15. Compute the Fisher information $I(\mu)$ for $\mathcal{N}(\mu, \sigma^2)$ with σ^2 known, and verify the sample mean $\bar{X}$ is efficient.
16. For the Poisson(λ), show $I(\lambda) = 1/\lambda$ and check that the MLE $\bar{X}$ attains the Cramér–Rao bound.
17. Write the exponential density in exponential-family form, identifying h, η, T, A .
18. Explain why a biased estimator is allowed to have variance below the CRLB.

6 Asymptotic normality of the MLE

We have a recipe (maximum likelihood), a quality scale (MSE, efficiency), a convergence guarantee (consistency), and a precision floor (Cramér–Rao). The final piece unifies them into the single most useful result in estimation theory: **for large samples, the MLE is approximately normal, centred at the truth, with variance equal to the Cramér–Rao bound.** This is what makes the MLE the default estimator of modern statistics — and what lets us attach error bars to it.

6.1 The statement

Worked example — the exponential MLE, sampled many times. Generate $n = 50$ exponential observations with true rate $\lambda = 2$, compute $\hat{\lambda} = 1/\bar{X}$, and repeat the whole experiment thousands of times. The histogram of the resulting $\hat{\lambda}$ values is a **bell curve**, centred very near 2, with spread about $\sqrt{\lambda^2/n} = 2/\sqrt{50} \approx 0.28$. The estimator is random, but its randomness has a precise, normal shape — and that shape is dictated entirely by the Fisher information.

Theorem 13 (asymptotic normality of the MLE) Under regularity conditions, the MLE $\hat{\theta}_n$ from n i.i.d. observations satisfies

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, 1/I(\theta)) \quad \text{as } n \rightarrow \infty,$$

where $\xrightarrow{d}$ denotes convergence in distribution and $I(\theta)$ is the per-observation Fisher information. Equivalently, for large n ,

$$\hat{\theta}_n \approx \mathcal{N}(\theta, 1/(nI(\theta))).$$

Read the three claims packed into this one line. **Consistency:** the mean is θ , so $\hat{\theta}_n \xrightarrow{P} \theta$. **Normality:** whatever the data’s distribution, the estimator’s sampling distribution is asymptotically a bell — the Central Limit Theorem of MAT31-2 acting on the score. **Efficiency:** the asymptotic variance is $1/(nI(\theta))$, **exactly the Cramér–Rao bound.** The MLE is, for large n , the best possible estimator — it sits on the floor.

Remark The deep reason is that the score $\ell'(\theta) = \sum_i (\partial/\partial\theta) \ln f(X_i; \theta)$ is a **sum of i.i.d. terms**, each with mean zero and variance $I(\theta)$. The CLT makes that sum asymptotically normal; a one-step Taylor expansion of the score around θ then transfers the normality to $\hat{\theta}_n$, with the Fisher information appearing as the conversion factor. The asymptotic normality of the MLE is the CLT in disguise.

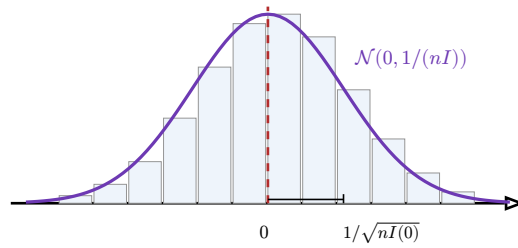


Figure 7: Asymptotic normality of the MLE. Across many samples, $\hat{\theta}_n$ is distributed approximately normally, centred at the true θ with standard deviation $1/\sqrt{nI(\theta)}$ — the square root of the Cramér–Rao bound. More data narrows the bell at rate $1/\sqrt{n}$.

6.2 The asymptotic variance and the standard error

The quantity $1/(nI(\theta))$ is the **asymptotic variance** of the MLE, and its square root $1/\sqrt{nI(\theta)}$ is the **asymptotic standard error** — the typical size of the estimation error. But it depends on the **unknown** θ . In practice we **plug in** the estimate, using $I(\hat{\theta}_n)$ in place of $I(\theta)$:

$$\text{se}(\hat{\theta}_n) = \frac{1}{\sqrt{nI(\hat{\theta}_n)}}.$$

This **plug-in principle** — substituting an estimate where a true value is needed — is justified because $\hat{\theta}_n \xrightarrow{P} \theta$ and I is continuous, so $I(\hat{\theta}_n) \xrightarrow{P} I(\theta)$. It is the bridge from the abstract theorem to a number we can actually report, and the foundation of the confidence intervals in the next chapter.

Common pitfall Asymptotic normality is a **large-sample** statement. For small n the sampling distribution of the MLE can be visibly skewed and the normal approximation poor — the same caution as the CLT in MAT31-2. The regularity conditions also genuinely matter: they fail for $\text{Unif}(0, \theta)$, whose MLE $\max_i X_i$ is **not** asymptotically normal (its rescaled error follows an exponential-type law, not a bell). Boundary-parameter models are the standard exception; do not apply the formula blindly.

Exercises

19. For $X \sim \text{Exp}(\lambda)$, write the asymptotic distribution of $\hat{\lambda}$, and give its asymptotic standard error in terms of $\hat{\lambda}$ and n .
20. For $\text{Poisson}(\lambda)$, state the asymptotic distribution of the MLE $\bar{X}$ and compare its asymptotic variance to the exact variance of $\bar{X}$.
21. Explain why the asymptotic-normality theorem does not apply to the uniform maximum, referring to the regularity conditions.

7 Asymptotic confidence intervals

Asymptotic normality is not merely beautiful — it is **operational**. It hands us, for free, a confidence interval for **any** parameter estimated by maximum likelihood, and a formula for **how much data** a target precision requires. This closes the loop opened in MAT31-2, where confidence intervals were available only for the mean.

7.1 Building the interval from the plug-in principle

Worked example — a confidence interval for a failure rate. From $n = 100$ lifetimes we compute $\hat{\lambda} = 0.5$ per hour. The asymptotic distribution is $\hat{\lambda} \approx \mathcal{N}(\lambda, \lambda^2/n)$, so the standard error, plugging in, is $\text{se} = \hat{\lambda}/\sqrt{n} = 0.5/10 = 0.05$. A 95% interval stretches 1.96 standard errors either side:

$$0.5 \pm 1.96 \times 0.05 = 0.5 \pm 0.098 = (0.40, 0.60).$$

We are 95% confident — **in the procedure’s sense** — that the true rate lies in this band.

Definition 14 (asymptotic confidence interval) For a parameter estimated by an asymptotically normal estimator with standard error $\text{se}(\hat{\theta}) = 1/\sqrt{nI(\hat{\theta})}$, an **asymptotic $(1 - \alpha)$ confidence interval** is

$$\hat{\theta} \pm z_{\alpha/2} \text{se}(\hat{\theta}),$$

where $z_{\alpha/2}$ is the standard-normal quantile with $P(Z > z_{\alpha/2}) = \alpha/2$. For 95%, $z_{0.025} = 1.96$; for 90%, 1.645; for 99%, 2.576.

The construction reuses everything: asymptotic normality supplies the **shape**, the Fisher information supplies the **width**, and the plug-in principle makes the width **computable**. The interpretation is exactly that of MAT31-2 — a property of the **procedure**, not a probability statement about θ .

Common pitfall The interval is a statement about the **method**, not the parameter. “95% confidence” means: **if we repeated the whole sampling-and-interval procedure many times, about 95% of the intervals would contain the true θ** . It does **not** mean “there is a 95% probability that θ is in this particular interval” — θ is a fixed number, either in or out. This is the same misreading flagged in MAT31-2, and it survives unchanged into the asymptotic setting.

Common pitfall These intervals are **approximate**, valid for **large n** . For small samples the coverage can be well below the nominal 95% — the interval is too short because the normal approximation overstates the precision. When n is small and the population is roughly normal with unknown variance, the **Student- t** interval of MAT31-2 is the honest correction; the asymptotic interval is what you reach for when n is comfortably large.

7.2 Determining the sample size

Because the half-width of the interval is $z_{\alpha/2}/\sqrt{nI(\theta)}$, we can **invert it**: demand a target half-width E and solve for the n that delivers it. This is study design — deciding **before collecting data** how much you need.

Proposition 15 (sample size for a target precision) To make the half-width of a $(1 - \alpha)$ confidence interval at most E , take

$$n \geq \frac{(z_{\alpha/2})^2}{E^2 I(\theta)}.$$

Since $I(\theta)$ depends on the unknown θ , use a pilot estimate or a worst-case (conservative) value.

Worked example — how many lifetimes for a rate to within 0.05? We want a 95% interval for the exponential rate λ with half-width $E = 0.05$, and a pilot study suggests $\lambda \approx 0.5$, so $I(\lambda) = 1/\lambda^2 = 4$. Then

$$n \geq \frac{(1.96)^2}{0.05^2 \times 4} = \frac{3.8416}{0.01} = 384.16,$$

so $n = 385$ observations. Note the **quadratic** cost: halving the half-width to 0.025 would quadruple the requirement to about 1537 — the same $\sqrt{n}$ law that governed every precision calculation since the CLT.

Remark The sample-size formula exposes the three levers of precision. You get a tighter interval by **collecting more data** ($n \rightarrow \text{up}$), **accepting lower confidence** ($z_{\alpha/2} \rightarrow \text{down}$), or **studying a more informative regime** ($I(\theta) \rightarrow \text{up}$). The first costs money, the second costs certainty, the third is often outside your control. The formula makes the trade explicit — which is why every well-designed experiment performs this calculation in advance.

Exercises

22. From $n = 64$ observations the MLE of a Poisson rate is $\hat{\lambda} = 4$. Using $I(\lambda) = 1/\lambda$, give a 95% asymptotic confidence interval for λ .
23. How many Bernoulli trials are needed to estimate p (worst case $p = 0.5$, so $I(p) = 4$) to within $E = 0.02$ at 95% confidence?
24. A 95% interval for a parameter has half-width 0.10 from $n = 200$. Roughly how large must n be to shrink the half-width to 0.04, all else equal?
25. Explain, using the plug-in principle, why we may write $I(\hat{\theta})$ in place of $I(\theta)$ in the standard error.

The arc of this course. Two recipes build estimators — **method of moments** (match sample moments to theoretical ones) and **maximum likelihood** (maximise the log-likelihood via the score equation, checking the second-order condition). Three yardsticks judge them — **bias**, **MSE** ($= \text{Var} + \text{Bias}^2$), and **efficiency**, balanced through the **bias–variance tradeoff**. Three asymptotic ideas explain **why they work** — **convergence in probability** and **almost sure convergence** give **consistency**; the **Fisher information** $I(\theta)$ sets the **Cramér–Rao floor** $1/(nI(\theta))$ on every unbiased estimator’s variance; and the **asymptotic normality of the MLE** shows it reaches that floor, $\hat{\theta} \approx \mathcal{N}(\theta, 1/(nI(\theta)))$. From that one fact flow **asymptotic confidence intervals** $\hat{\theta} \pm z_{\alpha/2}/\sqrt{nI(\hat{\theta})}$ and the **sample-size** formula $n \geq z_{\alpha/2}^2/(E^2 I(\theta))$ — the theoretical backbone for understanding when, and how well, estimators work.